

Trainer Debrief — Lab 33

Capstone Presentation and Defence

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Present and defend your capstone project to a simulated steering committee (A1-A5, K1, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-33-capstone-defence-rubric.xlsx

Root cause rigour carries the heaviest weight (20%) because it is what separates a Black Belt from an enthusiastic problem solver: anyone can propose a cause, a Black Belt proves it and disproves the rest. The two criteria candidates most often fail are measurement system validity (they skip the MSA, or run it and ignore the failure) and control (a control plan with no named owner and no reaction plan). Under challenge, 'we did not test that, and here is what it would take to test it' scores higher than a confident guess.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

You delivered within time, answered challenges with evidence from your appendix rather than assertion, and acknowledged genuine weaknesses with a stated mitigation.

Where this sits in the capstone

The capstone assessment. Present your project to a simulated steering committee of peers and the trainer, then defend the analysis under challenge. Defending your methodology under pressure is the skill that separates a Black Belt from a Green Belt.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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